

Reconstruction of patient medical history from medico-administrative databases

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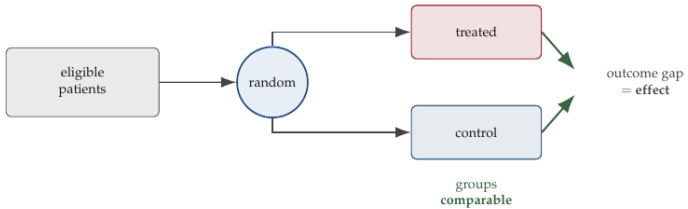
Why use medico-administrative databases ?

Clinical trials: the gold standard



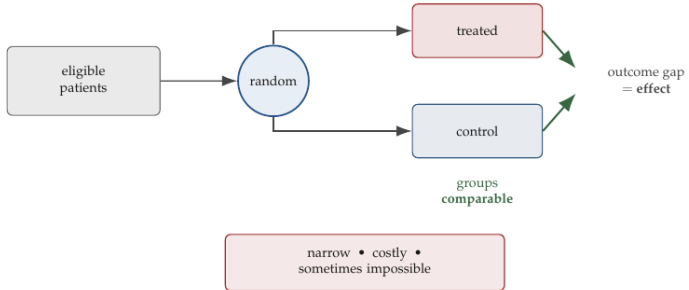
Randomisation is what guarantees an unbiased effect.

Clinical trials: the gold standard



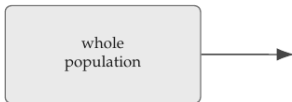
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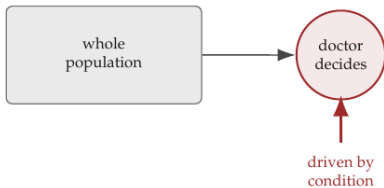


Randomisation is what guarantees an unbiased effect.

Real world data

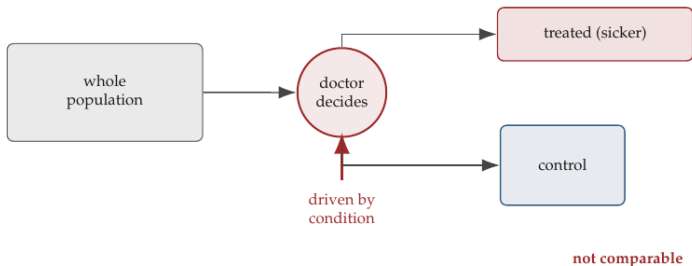


The link condition $\rightarrow$ treatment is exactly what randomisation removes.



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Real world data



The link condition $\rightarrow$ treatment is exactly what randomisation removes.

Clinical trials vs real world data

Randomised trial

- + unbiased *by design*
- + the causal gold standard
- small, selected, short
- not always feasible

Real-world data

- + large, representative
- + long follow-up, always there
- groups not comparable
- confounded

Adverse drug reactions

During clinical trials

- Only short term adverse drug reactions
- On a small subset of patients
- Without other drugs

Pharmacovigilance

- Adverse drug reactions are detected by the general practitioner.
- They are poorly detected.
- They are responsible for 8% of all hospitalizations.

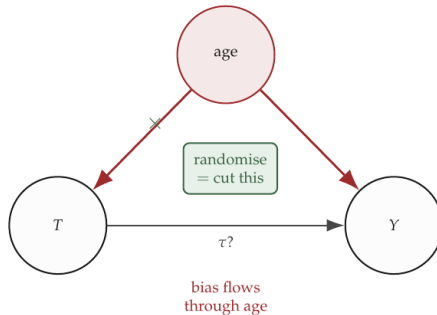
A massive medical database

- The medical history of 67 millions of patients
- For more than ten years
- Drug reimbursement and hospital stays

A claim database

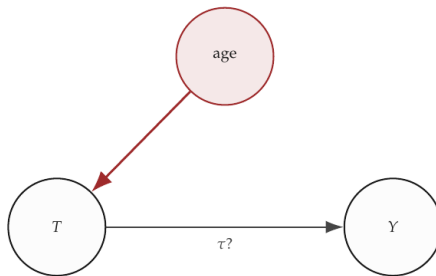
- The data was originally for administrative purpose
- It has built-in biases
- It has blind spots

The Causality problem



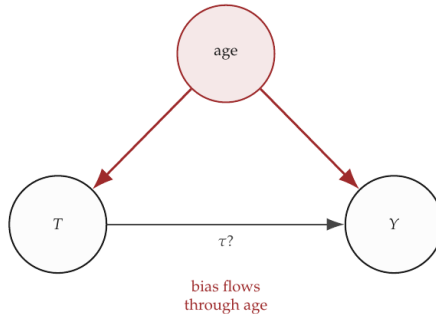
If older patients get the drug and also have worse outcome due to their age, the drug can look harmful even when it helps.

The causality problem



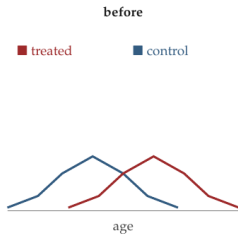
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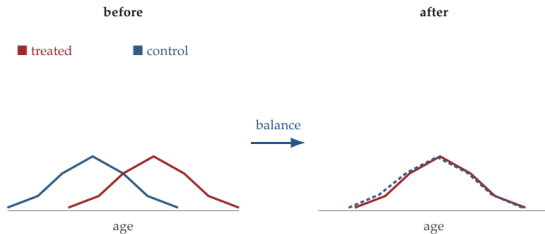
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Removing bias: balance the covariates distributions



The control age-density is reshaped until it matches the treated one.

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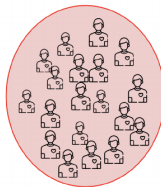


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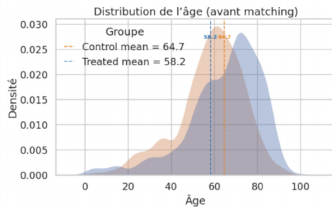
Matching



Treated group



Control group



Matching

Problem : the groups are **not exchangeable** due to **lack of randomization**



Solution : balance the groups by matching similar patients together

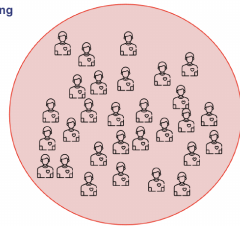
Matching procedure
(Distance-based matching)

> Select the patients from the control group that are **"similar"** to the ones in the treatment group

1:2 matching



Treated group



Control group

Matching

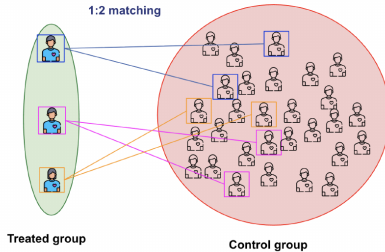
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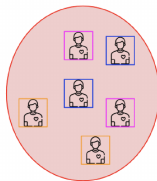
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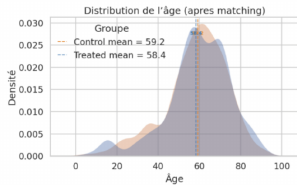
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Control group

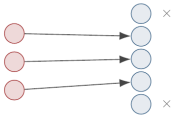


Matching or weighting methodology

Matching

pair each treated i with the control $m(i)$ nearest under a distance d

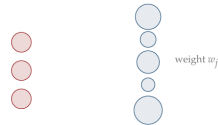
$$\min_m \sum_{i: T_i=1} d(X_i, X_{m(i)})$$



Weighting

keep every control; reweight them so their distribution matches the treated P_1 under the *same* d

$$\min_{w \geq 0} \mathcal{D}_d \left(P_1, \sum_{j: T_j=0} w_j \delta_{X_j} \right)$$



Both rest on one **patient distance** d (P_1 the treated distribution): matching pairs on d ; weighting matches the whole *distribution* under a discrepancy $\mathcal{D}_d$ built from d (energy distance / maximum mean discrepancy). Euclidean d on the means recovers $\|\bar{X}_1 - \sum_j w_j X_j\|$ – the open question is *which* d .

Example of covariates for distance

- Age
- Sex
- Charlson score (comorbidity index)

Adverse drug reaction analysis on the SNDS

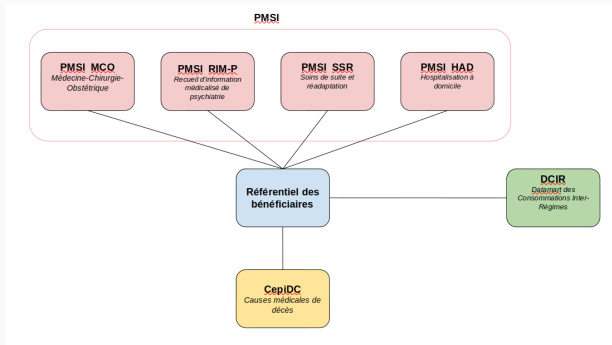
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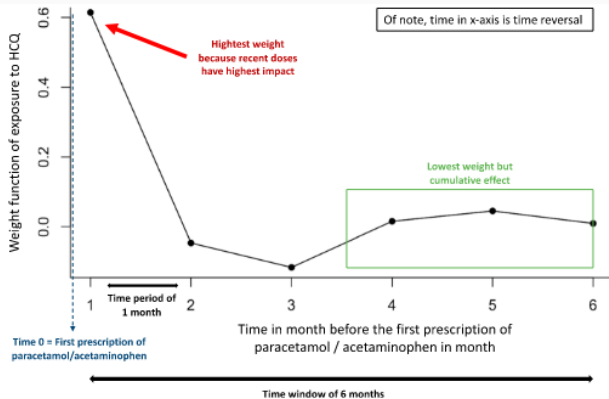
The SNDS



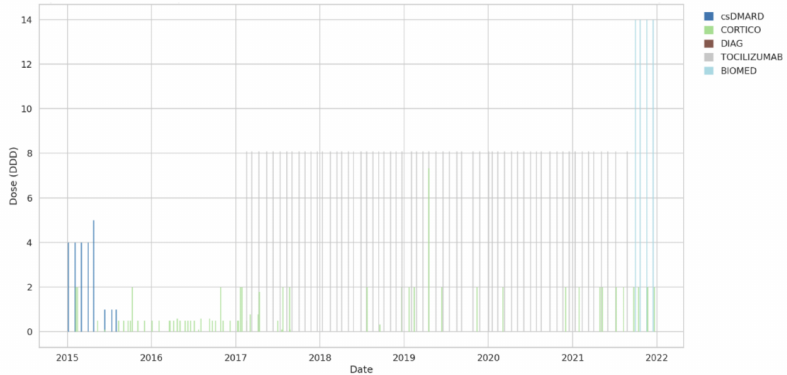
Type of studies interested in exposure reconstitution

- Adherence study
- Prevalence study
- Exposure effect on individual patient

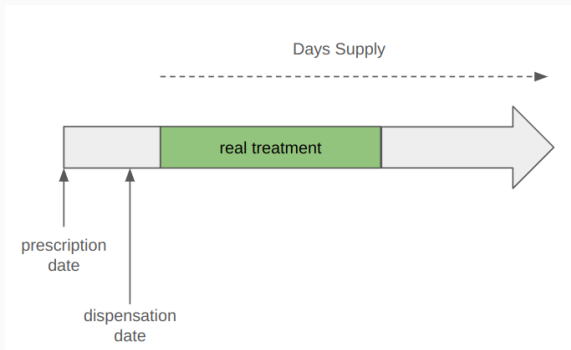
The WCE model for analyzing



Prescription in the SNDS



A prescription event



A prescription event

The real treatment of the patient

- Consumed Daily Dose (CDD)
- Treatment duration

If we assume that the patient take all his medication

- Consumed Daily Dose (PDD)
- Days Supply

The data we have

- Number of boxes
- Unit per box
- Dose per unit

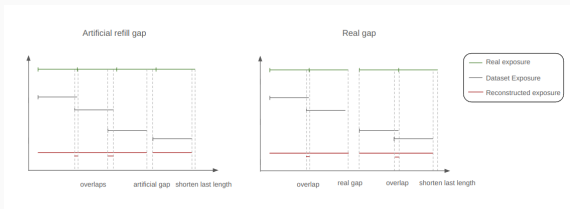
We need to estimate the Days Supply or the PDD

Limits on the type of treatment

- Patient must use all the prescribed medication
- No gap in prescription event
- Limited to chronic treatment

Reconstitution fo multiple prescription events

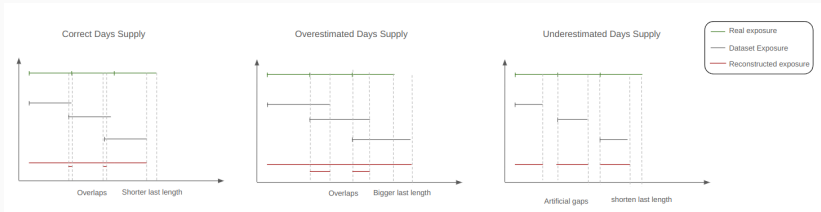
Challenges of event reconstitution



Early refill¹ and gap detection

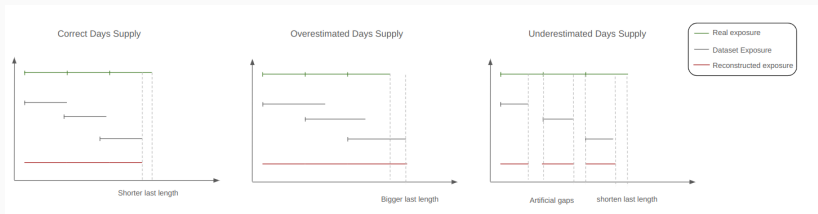
¹Carnahan, “Mini-Sentinel’s systematic reviews of validated methods for identifying health outcomes using administrative data”.

Challenges of event reconstitution



Overestimation and underestimation of the Days Supply

The different methods of episode reconstitution



Overlap Suppression

The different methods of episode reconstitution



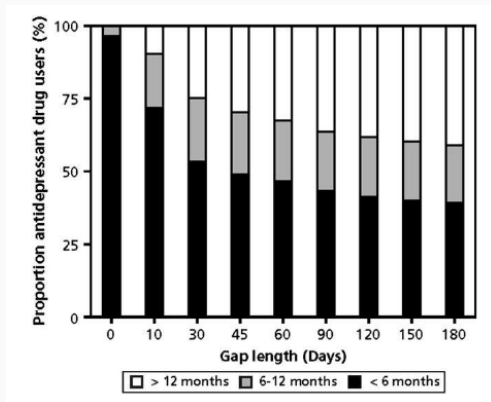
Grace period

The different methods of episode reconstitution



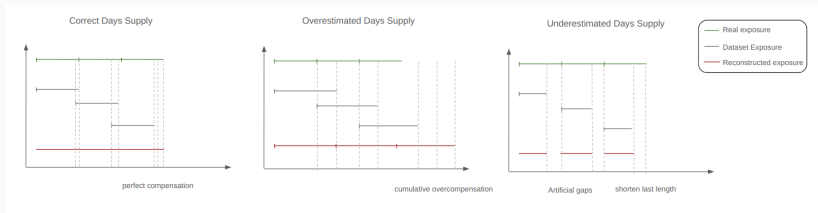
Grace period and overlap suppression

The different methods of episode reconstitution



Grace period duration influence

The different methods of episode reconstitution



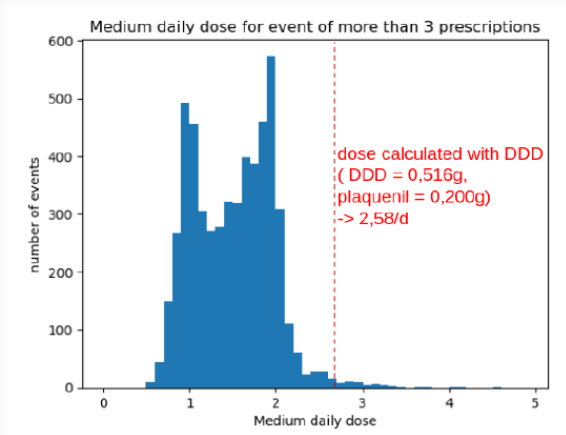
Episode reconstitution¹

¹Carnahan, "Mini-Sentinel's systematic reviews of validated methods for identifying health outcomes using administrative data".

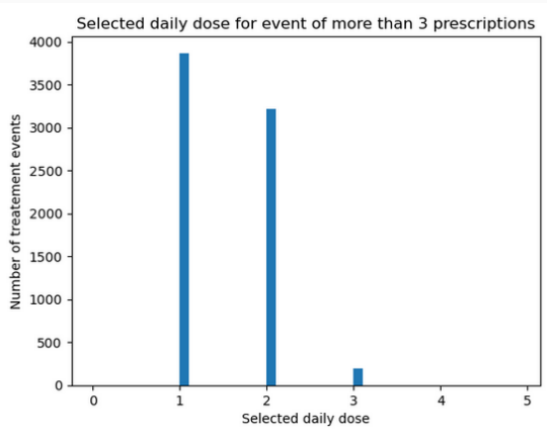
The different methods of episode reconstitution

	Underestimated Days Supply	Overestimated Days Supply	Early refill	Real Gaps detection	Computation complexity
Naive data	Artificial gaps Shorten last event	Big overlaps Bigger last event	Overlap Artificial gaps Shorten last event	Yes	
Overlap suppression	Artificial gaps Shorten last event	Bigger last event	Artificial gaps Shorten last event	Yes	Small
Grace period	Shorten last event	Big overlaps Bigger last event	Overlap Shorten last event	Susceptible to grace period definition	Small
Overlap + Grace period	Shorten last event	Bigger last event	Shorten last event	Susceptible to grace period definition	Small
Event reconstitution	Artificial gap Shorten last event	Cumulative error for each event, can be badly overestimated	Realistic representation	Yes	High
Event reconstitution + Grace Period	Shorten last event	Cumulative error for each event, can be badly overestimated	Realistic representation	Susceptible to grace period definition	High

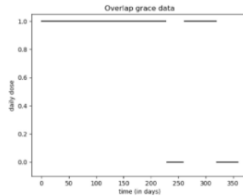
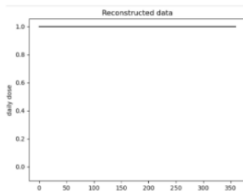
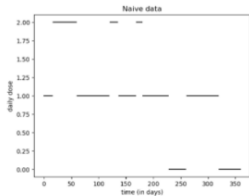
Exemple of Hydroxychloroquin



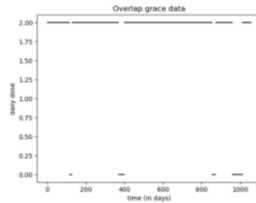
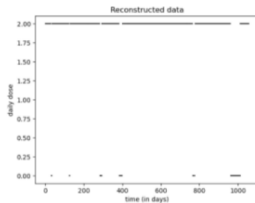
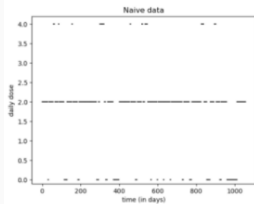
Example of Hydroxychloroquin



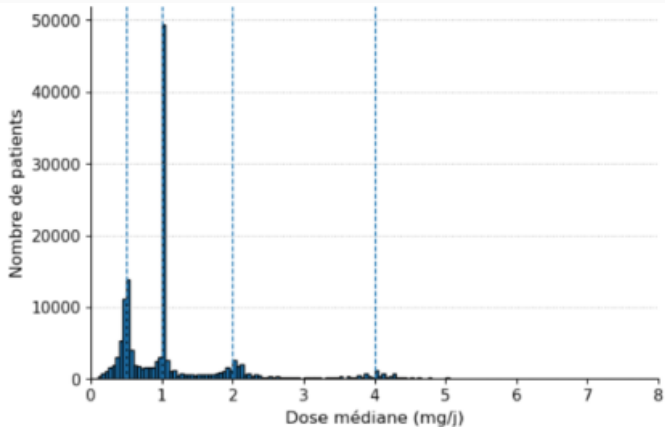
Reconstitution of a exposure



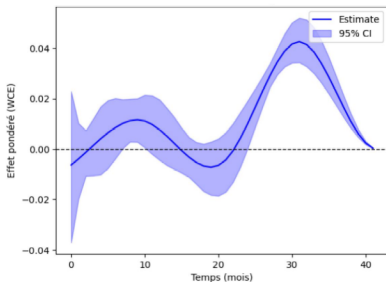
Reconstitution of an exposure



Atorvastatine adnd drug-induced hepaphitis



Atovastatine has an effect on drug-induced hepatic hepatitis



Hyperparamètres

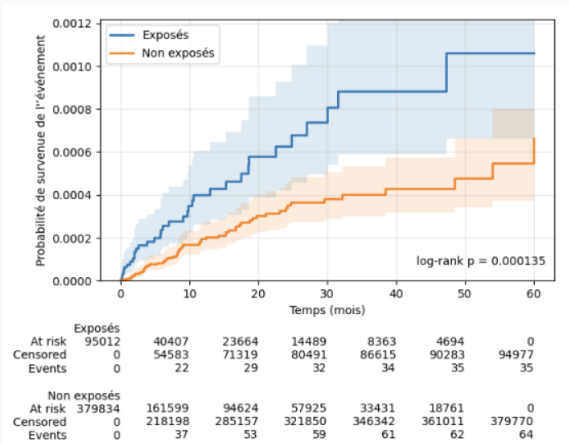
- Cutoff : 42
- Nknots : 3
- Constraint : Droite

Bootstrap

- Bootstrap : 1000
- Batchsize : 10

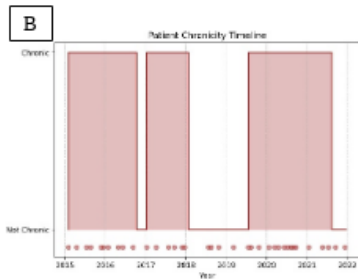
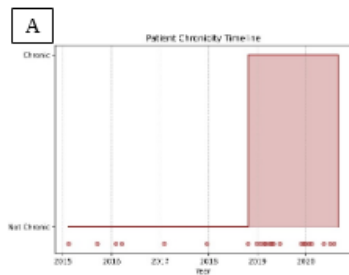
Hazard Ratio (IC95%) : 1,62 [1,54 ; 1,69]

Survival curve



A more complex simulation: corticoids

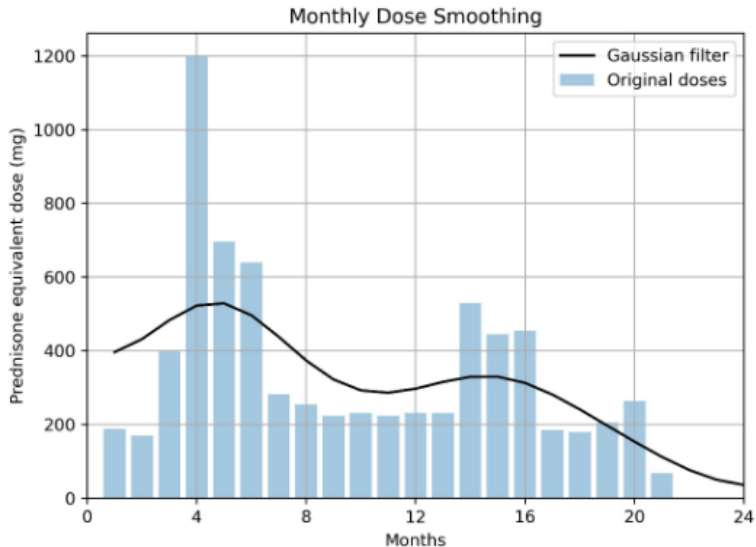
Definition of a chronic exposition



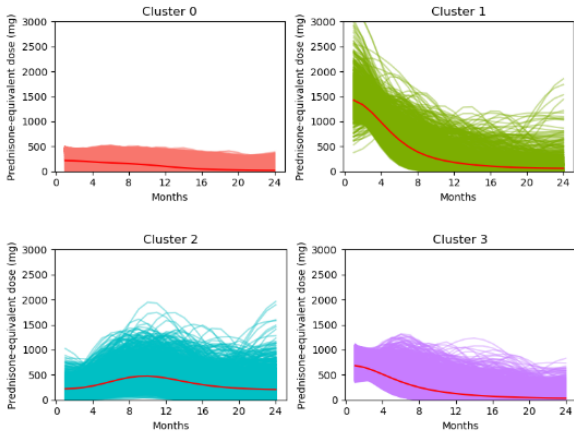
Patient matching

Variable	Chronic	Non-chronic
n	53,400	1,014,453
Female sex (%)	32,144 (60.1)	589,871 (58.1)
Age (mean (SD))	59.63 (17.55)	46.76 (18.61)
Weighted Charlson score (mean (SD))	1.37 (2.36)	0.38 (1.38)

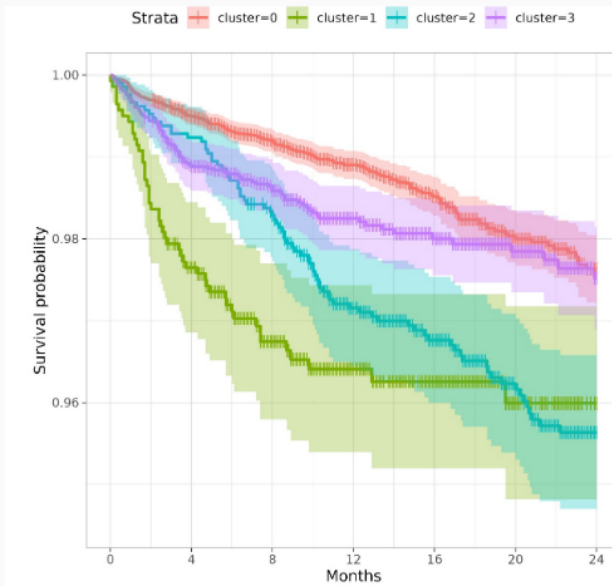
Modelling doses with Gaussian Smoothing



Modelling with Gaussian Smoothing



Survival of the different clusters



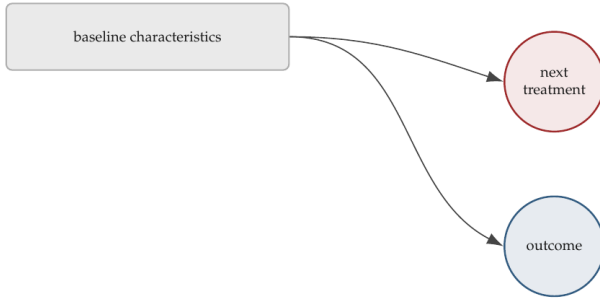
Interpretation on the clusters

Patient history for causal inference

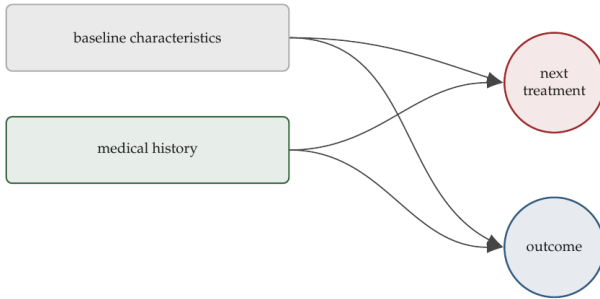
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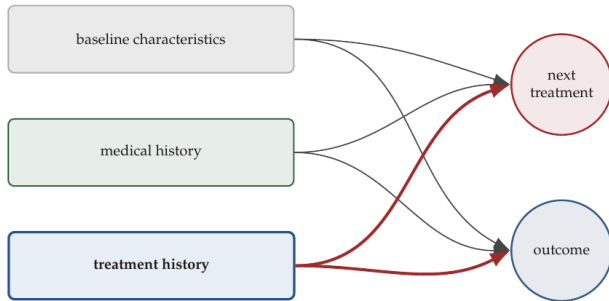
Three layers of confounding biases in patients



Three layers of confounding bias in patients



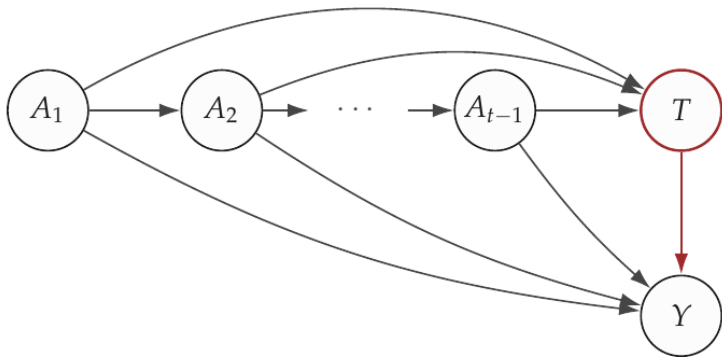
Three layers of counfounding bias in patients



strongest driver of the next decision – and the hardest to summarise

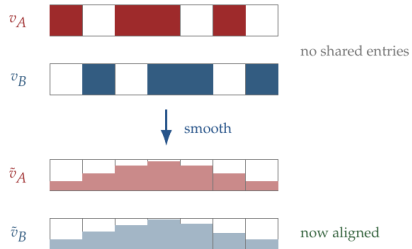
The causal structure

Causal structure



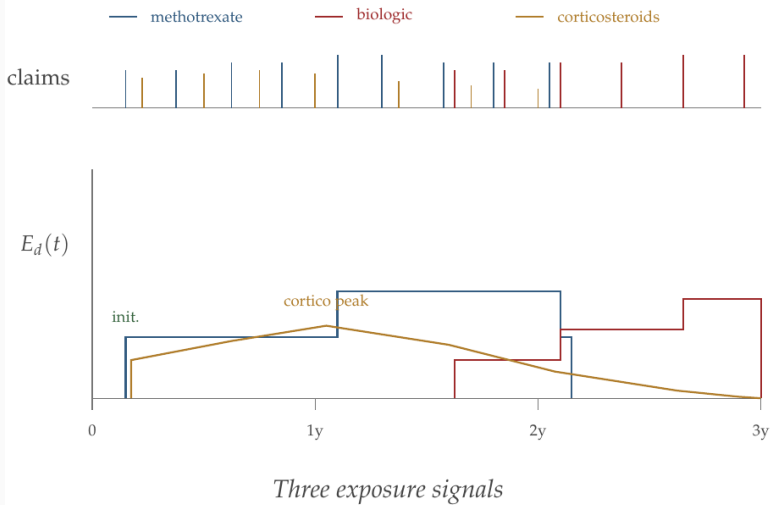
past treatments $A_{1:t-1}$ drive both T and Y

Event reconstruction



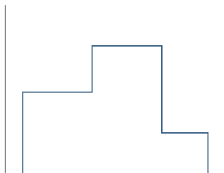
Different claim days $\Rightarrow$ disjoint one-hot vectors; after smoothing, the exposures coincide.

Event reconstruction



3 different smoothing functions

Step (coverage) function



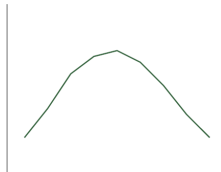
$$E_d(t) = \sum_k \frac{q_k}{\Delta_k} \mathbf{1}[t_k \leq t < t_k + \Delta_k]$$

Decay kernel



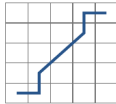
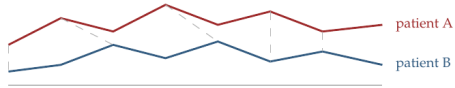
$$E_d(t) = \sum_k q_k e^{-(t-t_k)/\tau} \mathbf{1}[t \geq t_k]$$

Smoothing spline



$$E_d = \arg \min_f \sum_k (f(t_k) - q_k)^2 + \rho \int (f'')^2$$

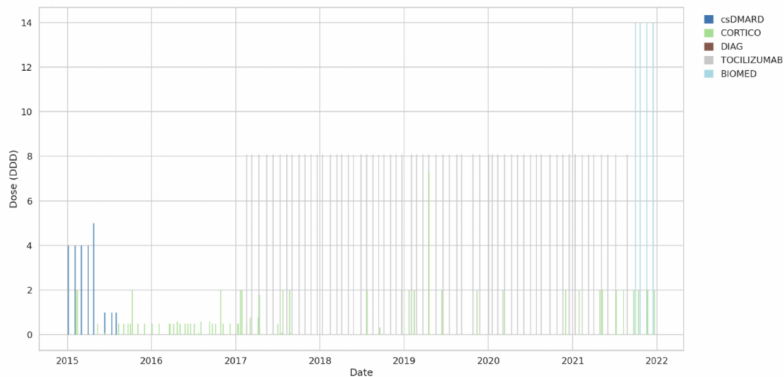
Alignement matching



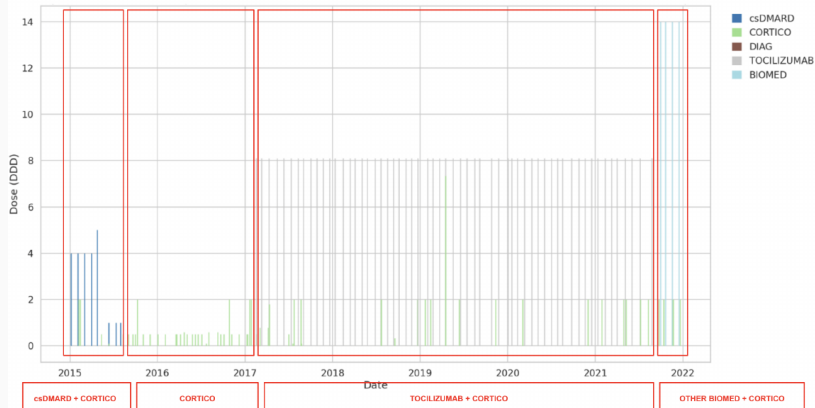
warping path

Alignment matches comparable phases at different times – but it is slow.

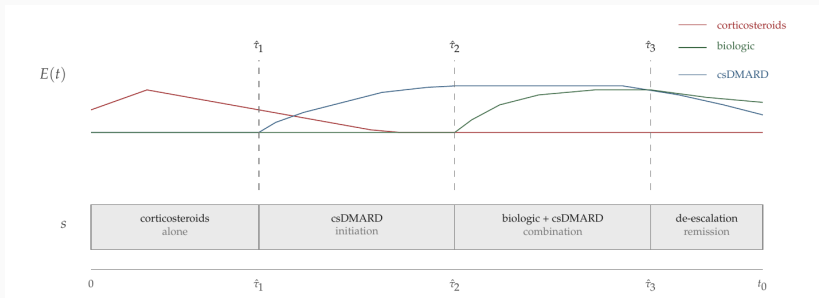
Prescription in the SNDS



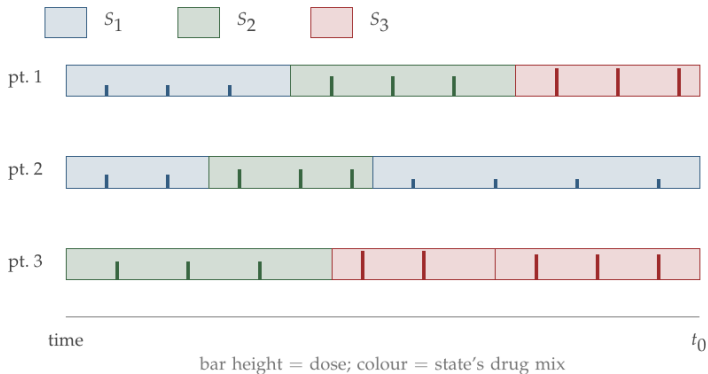
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Trajectory reconstruction

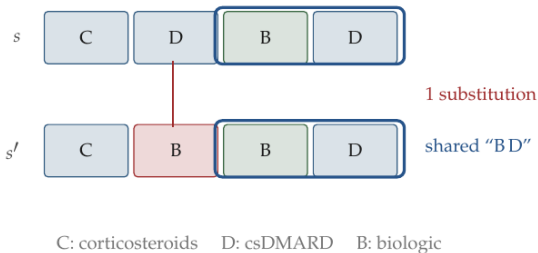


Patient trajectory



Unified states segment every patient and can generate synthetic histories.

Low dimension patient distance



Distances on sequences live in far lower dimension than the raw signal.